Yuanyuan Chen

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Education

2021 – 2026 **Doctor of Philosophy (Computer Science)**, *Nanyang Technological University*, Singapore
Advisor: Assoc. Prof. Han Yu

2005 – 2009 **B.S. in Mathematics and Applied Mathematics**, *Fuzhou University*, China

Research Interests

- Trustworthy Federated Learning
- Explainable Artificial Intelligence
- Generative Artificial Intelligence
- Graph Representation Learning

Work Experience

Dec. 2025 – Feb. 2026 **Research Fellow**, *Bake AI*, USA
I led the development of crust (**425+ GitHub stars**), an open-source security infrastructure for agentic AI systems that intercepts and blocks dangerous agent behaviors in real time.

Jul. 2025 – Nov. 2025 **Research Fellow**, *Deeper Semantics Lab, NTU*, Singapore

Oct. 2019 – Jun. 2025 **Research Fellow**, *NTU–WeBank Research Centre on FinTech*, Singapore
I contributed to multiple research projects at the Trustworthy Federated Ubiquitous Learning Research Centre.

Jun. 2019 – Sep. 2019 **Research Intern**, *WeBank*, Shenzhen, China
Federated learning is a popular distributed AI paradigm. During the internship, I implemented a federated-learning-based visual object detection system for operational safety, which received the AAAI 2020 Innovative Applications of Artificial Intelligence Award.

May 2016 – Feb. 2019 **System Architect**, *DeepIR Inc*, Xiamen, China
DeepIR delivers AI-powered services. My primary projects included an AI-based security module used at construction sites of the Beijing 2022 Winter Olympics venues, and a vehicle license plate recognition module used at the Hong Kong Customs checkpoint of the Hong Kong–Zhuhai–Macau Bridge.

Jul. 2009 – Apr. 2016 **Senior Software Engineer**, *ChinaSource Ltd*, Xiamen, China
Email service was one of the company's core businesses. Building on the open-source ecosystem, I developed a commercial mailing framework supporting the POP3, IMAP, and SMTP protocols.

Publications

Conference Proceedings

- Minghui Chen, Ruinan Jin, Wenlong Deng, **Yuanyuan Chen**, Zhi Huang, Han Yu, and Xiaoxiao Li, "Can Textual Gradient Work in Federated Learning?," *ICLR* 2025.
- Chen Chen, Ruizhe Li, Yuchen Hu, **Yuanyuan Chen**, Chengwei Qin, and Qiang Zhang, "Overcoming Catastrophic Forgetting by Exemplar Selection in Task-oriented Dialogue System," *ACL* 2024.
- Anran Li*, **Yuanyuan Chen***, Jian Zhang, Mingfei Cheng, Yihao Huang, Yueming Wu, Anh Tuan Luu, and Han Yu, "Historical Embedding-Guided Efficient Large-Scale Federated Graph Learning," *SIGMOD* 2024.
- **Yuanyuan Chen***, Zichen Chen*, Pengcheng Wu, and Han Yu, "FedOBD: Opportunistic Block Dropout for Efficiently Training Large-scale Neural Networks through Federated Learning," *IJCAI* 2023.

- **Yuanyuan Chen***, Zichen Chen*, Sheng Guo, Yansong Zhao, Zelei Liu, Pengcheng Wu, Chengyi Yang, Zengxiang Li, and Han Yu, "Efficient Training of Large-scale Industrial Fault Diagnostic Models through Federated Opportunistic Block Dropout," *IAAI* 2023.
- Zelei Liu*, **Yuanyuan Chen***, Yansong Zhao, Han Yu, Yang Liu, Renyi Bao, Jinpeng Jiang, Zaiqing Nie, Qian Xu, and Qiang Yang, "Contribution-Aware Federated Learning for Smart Healthcare," *IAAI* 2022.
- **Yuanyuan Chen**, Boyang Li, Han Yu, Pengcheng Wu, and Chunyan Miao, "HyDRA: Hypergradient Data Relevance Analysis for Interpreting Deep Neural Networks," *AAAI* 2021.
- Yang Liu, Anbu Huang, Yun Luo, He Huang, Youzhi Liu, **Yuanyuan Chen**, Lican Feng, Tianjian Chen, Han Yu, and Qiang Yang, "FedVision: An Online Visual Object Detection Platform Powered by Federated Learning," *IAAI* 2020.

Journal Articles

- Chuan Sun, **Yuanyuan Chen**, Xiaoli Tang, and Han Yu. 5555. " Efficient Federated Learning with Mean Block Difference-based Global Aggregation and Patience-based Local Training." *IEEE Transactions on Knowledge & Data Engineering,* no. 01 (April): 1–16.
- Rui Liu, **Yuanyuan Chen**, Anran Li, Yi Ding, Han Yu, and Cuntai Guan. 2024. "Aggregating intrinsic information to enhance BCI performance through federated learning." *Neural Networks* 172:106100.
- Zelei Liu*, **Yuanyuan Chen***, Yansong Zhao, Han Yu, Yang Liu, Renyi Bao, Jinpeng Jiang, Zaiqing Nie, Qian Xu, and Qiang Yang. 2023. "CAreFL: Enhancing smart healthcare with Contribution-Aware Federated Learning." *AI Magazine* 44 (1): 4–15.
- Hao Wang, Boyang Li, Haoming Zhong, Ahong Xu, Yingjie Huang, Jingfu Zou, **Yuanyuan Chen**, et al. 2022. "Smart Decision-Support System for Pig Farming." *Drones* 6 (12).
- Zelei Liu*, **Yuanyuan Chen***, Han Yu, Yang Liu, and Lizhen Cui. 2022. "GTG-Shapley: Efficient and Accurate Participant Contribution Evaluation in Federated Learning." *ACM Transactions on Intelligent Systems and Technology* 13, no. 4 (May).
- Yang Liu, **Anbu Huang**, Yun Luo, He Huang, Youzhi Liu, Yuanyuan Chen, Lican Feng, Tianjian Chen, Han Yu, and Qiang Yang. 2021. "Federated Learning-Powered Visual Object Detection for Safety Monitoring." *AI Magazine* 42, no. 2 (October): 19–27.

Awards & Honors

Academic Awards

2024 **Certificate of Accomplishment *for your outstanding performance and dedication towards the* ACM, Siggraph & Sigmobile NTU Student Chapter**, *NTU Association for Computing Machinery, Siggraph and Sigmobile*, Singapore

2023 **Deployed Application Award *for "Efficient Training of Large-scale Industrial Fault Diagnostic Models through Federated Opportunistic Block Dropout"***, *Association for the Advancement of Artificial Intelligence*

2022 **Deployed Application Award *for "Contribution-Aware Federated Learning for Smart Healthcare"***, *Association for the Advancement of Artificial Intelligence*

2020 **Innovative Applications of Artificial Intelligence Award *for "FedVision: An Online Visual Object Detection Platform Powered by Federated Learning"***, *Association for the Advancement of Artificial Intelligence*

Industry Honors

2025 **NVIDIA FLARE Summer of Code 2025**, *NVIDIA FLARE*

2023 **PyTorch Community Awards *for PyTorch Newcomer Nominee***, *The Linux Foundation*

Recognition for contributions to **PyTorch**, the world's leading open-source deep learning framework powering modern AI research and industry deployments.

2021 **Mars 2020 Helicopter Contributor**, *GitHub*

2020 **Arctic Code Vault Contributor**, *GitHub*

University Awards

2024 **5 Years Long Service Award *in appreciation for the service rendered to the university***, *Nanyang Technological University*, Singapore

2022 **Top 5 Most Popular Video Award *for Trust-based Open Collaborative Federated Learning***, *College of Engineering, Nanyang Technological University*, Singapore

Research Grants

2021–2025 **Participant, "TrustFUL: Trustworthy Federated Ubiquitous Learning"**, *Singapore Research Programme Grant (No. AISG2-RP-2020-019)*, S$6,883,200

Professional Services

Program Committee

2026 **The Fortieth AAAI Conference on Artificial Intelligence**, *AAAI-26*, Singapore

Academic Reviewer

2026 The Fortieth AAAI Conference on Artificial Intelligence

2025 IEEE Transactions on Neural Networks and Learning Systems

2024 IEEE Transactions on Neural Networks and Learning Systems

2024 International Journal of Intelligent Systems

2023 ACM Transactions on Software Engineering and Methodology

2022 IEICE Transactions on Fundamentals of Electronics, Communications and Computer Sciences

2022 IEEE Transactions on Neural Networks and Learning Systems

2020 The Thirty-Fourth AAAI Conference on Artificial Intelligence

Academic Conference Volunteer

2024 **The 2024 ACM Web Conference**, *WWW'24*, Singapore

2023 **The Thirty-Second International Joint Conference on Artificial Intelligence**, *IJCAI 2023*, Macao S.A.R.

Invited Talks & Presentations

2025 **NVIDIA FLARE Day APAC 2025**, *NVIDIA FLARE*, Online

2023 **Singapore ACM SIGKDD Symposium 2023**, *Singapore ACM SIGKDD Chapter*, Singapore

2018 **Invited Talk for Google Developer Groups**, *DevFest*, Xiamen, China

Projects

Award Winning Projects

2023 **FedOBD**

The Federated Opportunistic Block Dropout (FedOBD) approach was deployed for industrial fault diagnostic model training. It significantly reduces the communication overhead while maintaining model performance.

2022 **Contribution-Aware Federated Learning Framework**

The Contribution-Aware Federated Learning framework is designed and deployed for smart healthcare. It provides fair and explainable FL participant contribution evaluation in an efficient and privacy-preserving manner.

2020 **FedVision**

FedVision is a machine learning engineering platform to support the development of federated learning powered computer vision applications.

Other Projects

2017–2018 **AI Empowered Security System**
The system was deployed at the construction sites of the Beijing 2022 Winter Olympics venues.

2017 **Vehicle License Plate Recognition System for HZMB**
The system was developed in collaboration with partners for the Hong Kong–Zhuhai–Macau Bridge (HZMB) under the regulation and guidance of Hong Kong Customs.

2013–2016 **Commercial Mailing System**
A commercial mailing system was designed and developed on top of the open-source ecosystem. Common mailing protocols were supported as well as commercial extensions.

Membership

Professional Societies

2023–present **Member**, *Institute of Electrical and Electronics Engineers*

2022–present **Member**, *Association for Computing Machinery*

2022–present **Member**, *Association for the Advancement of Artificial Intelligence*

2016–2018 **Contributor**, *Association for Computing Machinery, Siggraph & Sigmobile NTU Student Chapter*

Engineering Organizations

2024–present **Member**, *ONNX*
ONNX is an open format built to represent machine learning models.

2020–present **Outside Collaborator**, *PyTorch*
An AI framework.

2024–present **Contributor**, *Protocol Buffers*
Protocol Buffers are language-neutral, platform-neutral extensible mechanisms for serializing structured data.

2024–present **Contributor**, *NVIDIA NVFlare*
NVIDIA FLARE is a domain-agnostic, open-source, and extensible SDK for Federated Learning.

2023–present **Contributor**, *Hugging Face*
The platform where the machine learning community collaborates on models, datasets, and applications.

2021–present **Contributor**, *NVIDIA DALI*
The NVIDIA Data Loading Library (DALI) is a GPU-accelerated library for data loading and pre-processing to accelerate deep learning applications.

2018–present **Contributor**, *OpenCV*
OpenCV is the world's biggest computer vision library.

2018–present **Contributor**, *Apache Thrift*
Thrift is a lightweight, language-independent software stack for point-to-point RPC implementation.

2018–present **Contributor**, *Faiss*
Faiss is a library for efficient similarity search and clustering of dense vectors.